

Reducing CNOT Count of MCX Decomposition

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June 2026

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1 Introduction

Gate decomposition, or the act of breaking down a complex gate involving multiple controls or targets into simpler gates, is necessary to simulate complex circuits on NISQ devices.

The MCX (multi-control X) gate is one of the most important ingredients in gate decomposition. Finding efficient ways to decompose the MCX is thereby essential to reducing the depth and 2-qubit count of quantum circuits.

Depending on whether there are ancilla qubits in the circuit, we decompose MCX with different methods. With no ancilla in an n -qubit circuit, our MCX has $(n - 1)$ controls, and the best known decomposition method is $\Theta(n^2)$ [1]. However, when there is a single ancilla in the circuit (our MCX has $(n - 2)$ controls), we can use a $\Theta(n)$ method for decomposition. The current upper bound calculated for this method is $16n - 40$ [2], and this bound can be further reduced by 12 to $16n - 52$ by replacing some toffolis (2-control X gates) with their approximate versions. In this report we will illustrate how to do that.

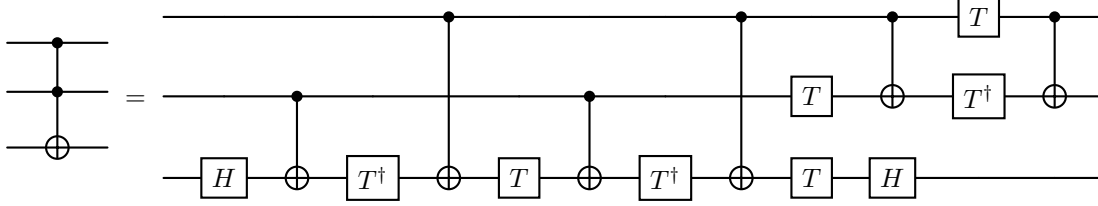
Moreover, we will share some insights on how to engineer approximate toffolis with specific phases and what versions of approximate toffolis to use in different circuits.

2 Gate Decomposition Methods

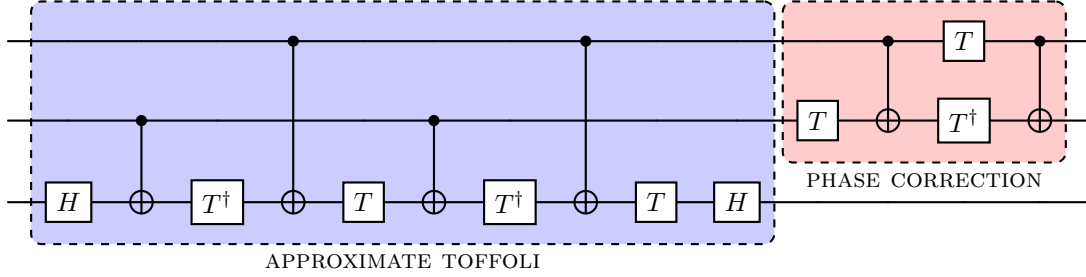
2.1 The Decomposition of a Toffoli

2.1.1 Exact Decomposition of a Toffoli

The standard textbook implementation of the toffoli gate is as follows [3]:



A little intuition for this construction is to see this circuit as composed of an *approximate toffoli* and a *phase correction* part.



The *approximate toffoli* part has T and $T^\dagger$ gates sandwiched between CNOTs, and this sandwich structure takes advantage of an important property of rotation and phase gates.

This important property is detailed in [1]:

1. $\sigma_x \cdot R_y(\theta) \cdot \sigma_x = R_y(-\theta)$
2. $\sigma_x \cdot R_z(\theta) \cdot \sigma_x = R_z(-\theta)$

Conjugating R_y and R_z rotation gates with σ_x gates flips the direction of rotation. T gates are equivalent to R_z gate with a global phase:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix} = e^{i\frac{\pi}{8}} \begin{pmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{pmatrix} = e^{i\frac{\pi}{8}} R_z\left(\frac{\pi}{4}\right)$$

$$T^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\frac{\pi}{4}} \end{pmatrix} = e^{-i\frac{\pi}{8}} \begin{pmatrix} e^{i\frac{\pi}{8}} & 0 \\ 0 & e^{-i\frac{\pi}{8}} \end{pmatrix} = e^{-i\frac{\pi}{8}} R_z\left(-\frac{\pi}{4}\right)$$

Then conjugating T with σ_x is equivalent to constructing $e^{i\frac{\pi}{4}} T^\dagger$:

$$\sigma_x \cdot T \cdot \sigma_x = e^{i\frac{\pi}{8}} \cdot R_z\left(-\frac{\pi}{4}\right) = e^{i\frac{\pi}{8}} \cdot (e^{i\frac{\pi}{8}} \cdot T^\dagger) = e^{i\frac{\pi}{4}} T^\dagger$$

and equivalently, conjugating $T^\dagger$ with σ_x is equivalent to constructing $e^{-i\frac{\pi}{4}} T$

$$\sigma_x \cdot T^\dagger \cdot \sigma_x = e^{-i\frac{\pi}{8}} \cdot R_z\left(\frac{\pi}{4}\right) = e^{-i\frac{\pi}{8}} \cdot (e^{-i\frac{\pi}{8}} \cdot T) = e^{-i\frac{\pi}{4}} T$$

Therefore, when both controls are active, the activated CNOTS that conjugate the $T^\dagger$ gates transform them into $e^{-i\frac{\pi}{4}} T$ gates. Then the part between the hadamards is equivalent to CCZ up to a phase:

$$e^{-i\frac{\pi}{4}} T \cdot T \cdot e^{-i\frac{\pi}{4}} T \cdot T = -i \cdot Z$$

Since conjugating a Z gate with H 's is equivalent to a σ_x gate, this is equivalent to having the target gets acted upon by a toffoli gate with a $-i$ phase. The extra $-i$ phase is why we call this part the *approximate toffoli* part.

To turn this approximate toffoli into an exact toffoli, we want to get rid of the extra $-i$ phase, which gets activated when both controls are 1. To do that we can apply a CS (controlled S) gate to the two controls, which applies a i phase when both qubits are 1. This way, the total phase of the circuit cancels: $i \cdot (-i) = 1$. The highlighted *phase correction* part is the decomposition of this CS gate.

It is also not difficult to verify that when one or both of the controls are zero, nothing happens to the target and both controls due to the symmetry of this circuit construction.

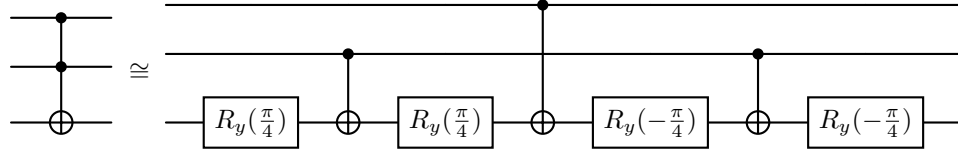
This 6-CNOT implementation of the toffoli has been proven to be the optimal exact decomposition [4]. However, there are cases where we don't need to implement the exact version of a toffoli. In fact, it suffices to implement their approximate version, which we introduce in the following section.

2.1.2 Approximate Decomposition of a Toffoli

Margolus showed that a phase-modified toffoli (approximate toffoli) can be realized with only 3 CNOTs, and Song and Klappenecker proved that they are optimal [5]. Since 2-qubit entangling gates such as CNOTs are expensive operations on NISQ devices, replacing exact toffolis with their approximate versions whenever possible could significantly increase circuit implementability.

There are many ways to construct approximate toffolis, and each method incurs a different phase. We will illustrate some methods below and demonstrate their differences.

Below shows [1]'s implementation of an approximate toffoli:



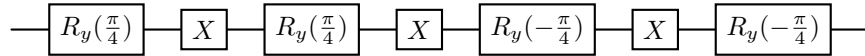
There's two things we can notice about this construction:

1. Single qubits gates only include rotation gates and their inverses.
2. These single qubit rotations sandwiches CNOTs in the middle.

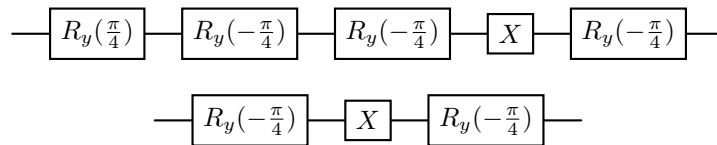
This structure is very similar to the sandwich structure we've seen previously, and the intuition for this construction is that we are still taking advantage of this fact:

1. $\sigma_x \cdot R_y(\theta) \cdot \sigma_x = R_y(-\theta)$
2. $\sigma_x \cdot R_z(\theta) \cdot \sigma_x = R_z(-\theta)$

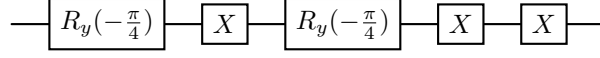
Therefore we can examine what operations are applied to the target qubit when both controls are active:



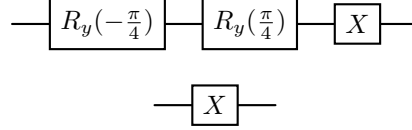
Using our X-conjugation trick we can now turn $\sigma_x \cdot R_y(\frac{\pi}{4}) \cdot \sigma_x$ into $R_y(-\frac{\pi}{4})$, and we observe that $R_y(\frac{\pi}{4})$ and $R_y(-\frac{\pi}{4})$ cancels out to identity:



Now we want to apply the X-conjugation trick again to cancel out the rest of the rotations, but we are missing a σ_x gate to the right to flip the direction of rotation. Since $\sigma_x \cdot \sigma_x = I$ there is nothing stopping us from adding two consecutive X gates to the right:



Now we apply our X-conjugation trick again:



Therefore, we can see that when both controls are active, an X gate is applied to the target qubit, which is exactly what a toffoli does.

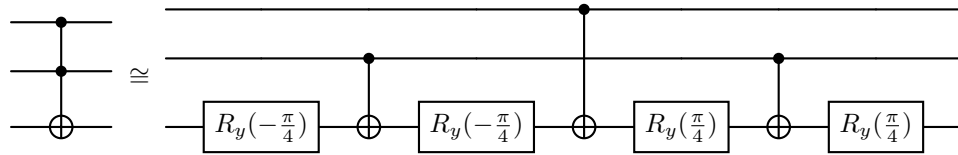
As with our approximate toffoli in the previous section, this approximate toffoli also incurs a phase. Here we'll use the little-endian convention ($|q_2q_1q_0\rangle$)

Where does it give an extra phase? Let's compute its matrix and see:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

We can see that the 5th column has a phase, so it adds a -1 phase to the state $|101\rangle$. The other parts are identical to a typical toffoli.

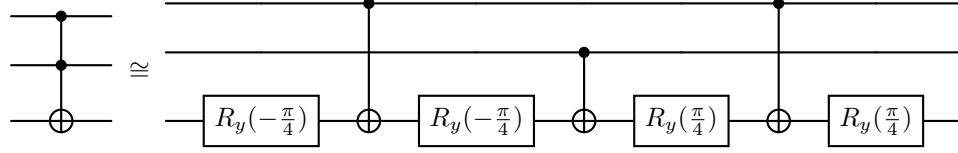
The interesting thing here is that you can engineer the phase to be on specific input states by playing around with the order of gates. Suppose we put $R_y(-\frac{\pi}{4})$ first instead of $R_y(\frac{\pi}{4})$, then we'll have:



$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Now the phase is on $|001\rangle$.

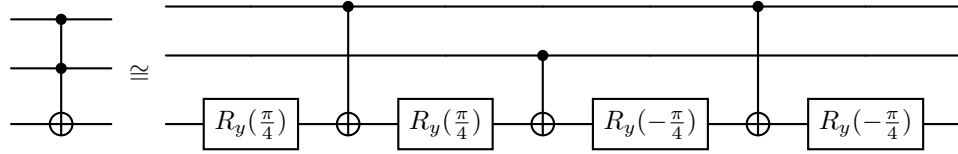
Another possible way to engineer the phase comes from [2].



$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The phase is now on $|010\rangle$.

If we swap the order of $R_y(-\frac{\pi}{4})$ and $R_y(\frac{\pi}{4})$ again we have:



$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The phase is now on $|110\rangle$.

Here, while there seems to be 4 possibilities where the phase could land on, namely $|101\rangle, |001\rangle, |010\rangle$ and $|110\rangle$, they belong to 2 groups. One group is of the form $q_0 = 1, q_1 = 0$ and the other is of the form $q_0 = 0, q_1 = 1$.

Depending on the specific structure of the circuit, most of the times only 1 group can be used. We will demonstrate this later on.

The approximate toffoli comes in handy when it comes to decomposing MCX gate with many controls when the circuit has ancillas. This case is introduced in the following section.

2.2 MCX with $\geq (\text{num_controls} - 2)$ ancillas

The most important thing to gain from this section is:

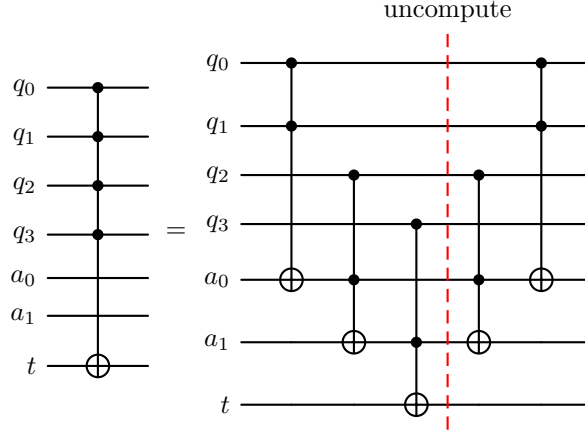
1. the toffoli-chain method of decomposing a MCX
2. how toffolis not acting on the target can be replaced by their approximate versions

We build up the intuition for the toffoli-chain method by starting from the case where all ancillas are clean.

2.2.1 When you have clean ancillas:

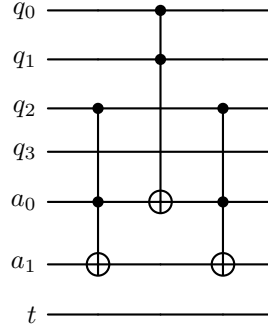
When all ancillas are clean, we can use the classical AND tree method to compute the result of 2 qubits onto an ancilla qubit ($a_0 = q_0 \wedge q_1$). Then, we compute the result of the ancilla qubit (which now contains the value of $q_0 \wedge q_1$) and another qubit (q_2) onto another ancilla ($a_1 = a_0 \wedge q_2 = (q_0 \wedge q_1) \wedge q_2$). This goes on. At the end of this chain, we can see that the target contains the value of $((q_0 \wedge q_1) \wedge q_2) \wedge q_3$, which is exactly what a MCX would do. This process is shown by the first 3 gates in the figure below.

After computing the result onto the target, we must uncompute the ancillas (in other words, reset the ancillas to their original value), and this is done by the last 2 gates in the figure below.



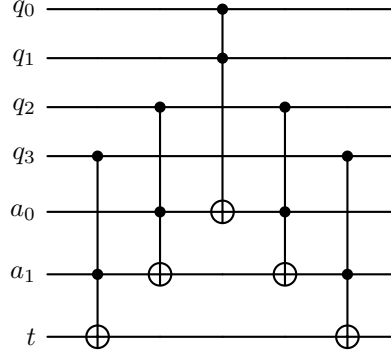
2.2.2 When you have dirty ancillas:

When the ancillas are dirty the situation becomes a bit annoying, because now the initial value of the ancillas leaks into the final target value. The trick is to construct a circuit structure that allows you to compute $q_0 \wedge q_1 \wedge q_2 \wedge q_3$ but is agnostic of the initial value of the ancillas. The structure is as follows:



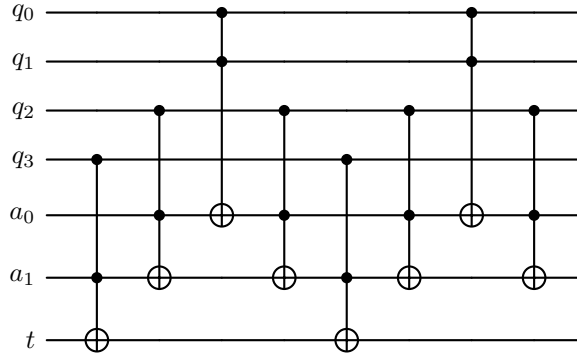
A little bit of intuition for this construction is that a_0 only flips when $q_0 \wedge q_1 = 1$. Likewise a_1 only flips when $a_0 \wedge q_2 = (q_0 \wedge q_1) \wedge q_2 = 1$, so this structure would still allow you to compute $((q_0 \wedge q_1) \wedge q_2) \wedge q_3$ onto the target.

However, what's special about this symmetric structure is that it cancels out the initial value of the ancillas. Notice that with this structure, the target (which is a_1 now), gets the value of $q_0 \wedge q_1 \wedge q_2$ because the $q_2 \wedge a_1$ part appears twice and cancels out. Therefore, with this series of cascade structure we can make it so that t gets the value of $q_0 \wedge q_1 \wedge q_2 \wedge q_3$ and is agnostic of the initial ancilla values, which is exactly what we want.



This is how we can compute t .

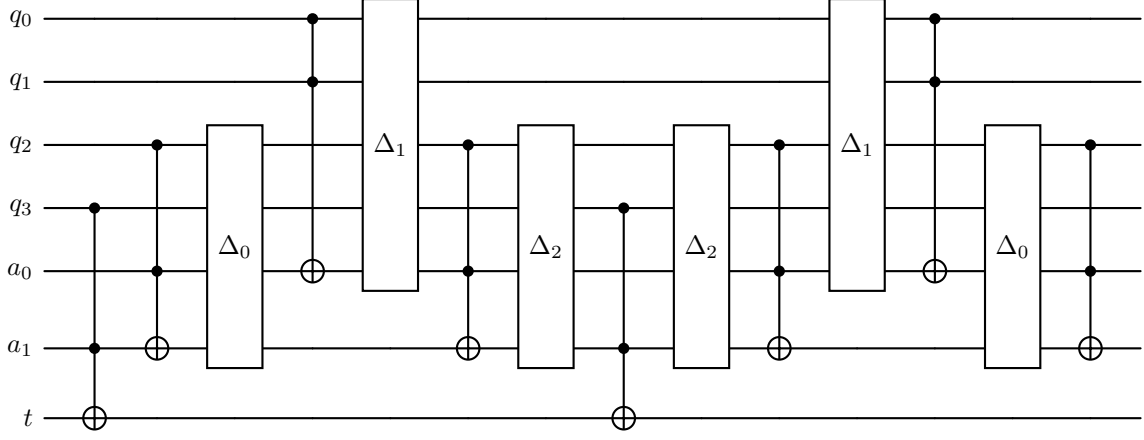
Then we just need to add the uncompute part:



Therefore, any MCX-gate with k controls can be simulated by a chain of x toffolis, where x is given by the following calculation: each chain circuit can be split into four-parts: target-toffoli (a toffoli acting on the target), reset part, target toffoli, reset part. The reset part has 1 toffoli in the middle and $(k-3)$ pairs of toffolis, so each reset part has $1+2(k-3)$ toffolis, and both reset parts have $2+4(k-3) = 4k-12+2 = 4k-10$ toffolis. Counting the 2 target toffolis, the total toffoli count is $4k-10+2 = 4k-8$.

If we simulate every toffoli with an exact toffoli (which costs 6 cnots), the CNOT count of a k control MCX is $6(4k-8) = 24k-48$. However, people have pointed out that all toffolis in the reset part can be replaced with approximate ones that only cost 3 CNOTS per toffoli [1] [2]

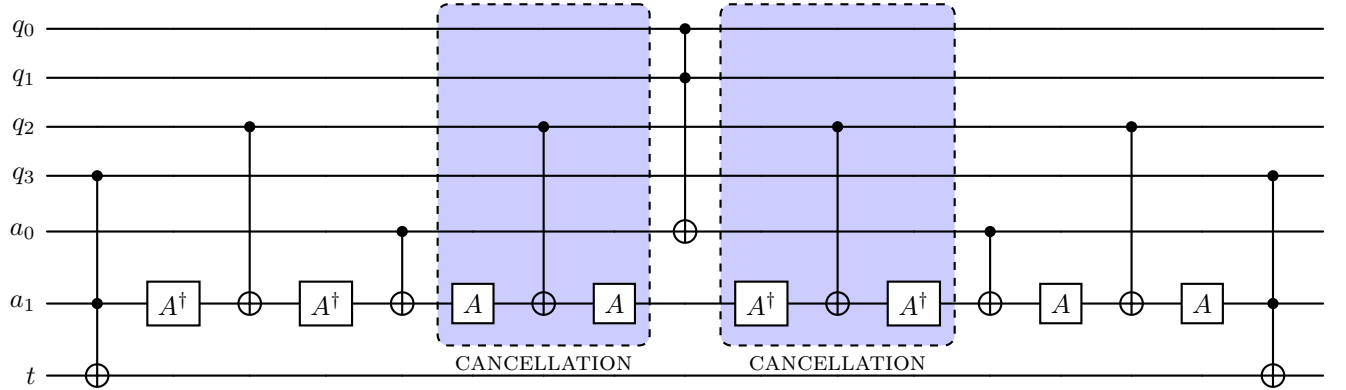
The proof that toffolis not acting on the target can be replaced by approximate ones is as follows (Lemma 8 [2]):



The main idea is:

1. A approximate toffoli gate can be thought of as a toffoli gate composed with a diagonal gate that adds a phase when the input state is $|101\rangle$
2. If a diagonal is acting on the controls of a gate, it commutes with the gate (Lemma 7 [2]). Henceforth, the Δ_2 's and the Δ_1 's cancel out.
3. Moreover, we can think of the gates between the 2 Δ_0 's as a single composite gate. This gate only act on the target & does nothing to the other qubits, so we can view the Δ_0 's as acting on the control qubits of the composite gate in the middle. Hence the Δ_0 's cancel out as well.
4. Therefore, all toffolis not acting on the target can be replaced with their approximate counterparts, thereby significantly reducing our total CNOT count.

Moreover, further cancellations can happen here [2]:

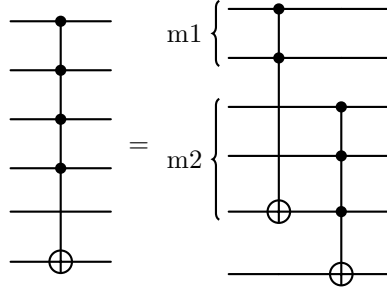


The last 3 gates before the toffoli can be canceled with its mirror gate group to the right of the last toffoli. Hence, in the reset group, there's a toffoli in the middle contributing 3 CNOTs, and there are $k - 3$ pairs of toffolis contributing 4 CNOTs per pair, therefore yielding a CNOT contribution of $3 + 4(k - 3)$. As a consequence, 2 reset parts plus 2 exact toffolis yield a CNOT count of $6 + 8(k - 3) + 12 = 18 + 8k - 24 = 8k - 6$ [2].

2.3 MCX With One Ancilla

For MCX with one ancilla, you want to break the $(n - 2)$ controlled MCX gates into smaller pieces. Specifically, the idea is to let $m_1 = \lfloor \frac{n}{2} \rfloor - 1$ and $m_2 = \lceil \frac{n}{2} \rceil$ so that $m_1 + m_2 = n - 1$.

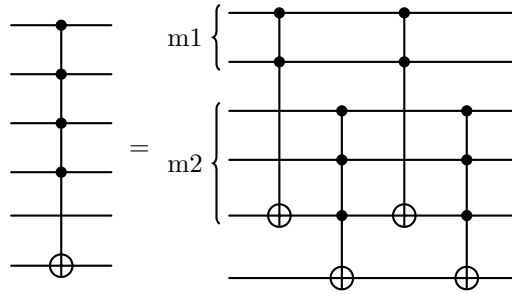
2.3.1 When you have a clean ancilla:



The intuition here is simple. After the first MCX, the ancilla carries $q_0 \wedge q_1$. The next MCX gate then computes that onto the target $t = t \oplus (q_0 \wedge q_1) \wedge q_2 \wedge q_3$.

2.3.2 When you have a dirty ancilla:

Source: Lemma 7.3 [1]



The annoying thing here is that the ancilla is not 100% 0, so if you just do the clean ancilla trick you end up with $t = t \oplus (a \oplus q_0 \wedge q_1) \wedge q_2 \wedge q_3 = t \oplus q_0 \wedge q_1 \wedge q_2 \wedge q_3 \oplus a \wedge q_2 \wedge q_3$.

So there is an extra $a \wedge q_2 \wedge q_3$ term you want to get rid of. The trick to doing so is to add a $a \wedge q_2 \wedge q_3$ term to the target because $a \wedge q_2 \wedge q_3 + a \wedge q_2 \wedge q_3 = 0$. The way you do that is just to apply the circuit again.

When you do the m_1 controlled MCX gates on the ancilla again you get $a = (a \oplus q_0 \wedge q_1) \oplus q_0 \wedge q_1 = a$. Then, when you apply the m_2 controlled MCX gates again you get $t \oplus q_0 \wedge q_1 \wedge q_2 \wedge q_3 \oplus a \wedge q_2 \wedge q_3 \oplus a \wedge q_2 \wedge q_3 = t \oplus q_0 \wedge q_1 \wedge q_2 \wedge q_3$.

The genius of dividing up the MCX this way is that it allows you to use the toffoli-chain method. More specifically, the MCX with m_1 controls can have m_2 ancilla qubits, and the MCX with m_2 controls can have m_1 ancilla qubits.

What is the CNOT count of this construction? If you construct each part with the approximate toffoli chain method, you'll have $16n - 40$ CNOTS as your upper bound. The calculation goes as follows:

$$count = 2[8m_1 - 6 + 8m_2 - 6] = 16(m_1 + m_2) - 24 = 16(n - 1) - 24 = 16n - 40$$

3 Improvements

3.1 Upper Bound of MCX Decomposition can be Reduced

A multi-controlled X gate is built out of many Toffoli gates. An exact Toffoli costs 6 CNOTs; a cheaper relative-phase Toffoli does the same flip with only 3, but on one particular input it leaves behind a stray -1 phase. So every time we trade an exact Toffoli for a cheap one we save three CNOTs but incur a phase debt. The entire question of this section is: *can we arrange for those phase debts to cancel?* We will show that for the Toffolis that target the ancilla qubit they can be made to cancel, but only if the phase of approximate

toffolis have a certain activation state, and that cashing in four such discounts tightens the known $16n - 40$ bound [6] to $16n - 52$.

3.2 Why the debt has a chance of cancelling

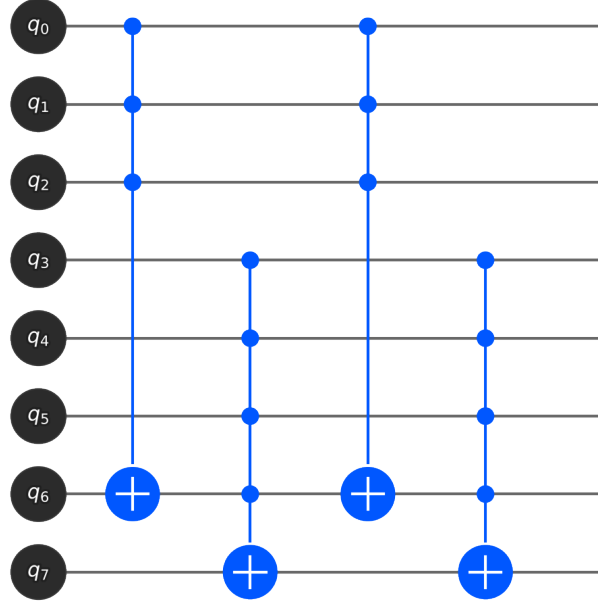


Figure 1: MCX gate decomposed.

The construction we start from [1, 2] realises the $(n - 2)$ -controlled σ_x with one borrowed (possibly dirty) ancilla as a four-piece product

$$C_{n-2,n}(\sigma_x) = (m_1\text{MCX}) (m_2\text{MCX}) (m_1\text{MCX}) (m_2\text{MCX}),$$

in which the two $m_1\text{MCX}$ blocks are *identical*, and likewise the two $m_2\text{MCX}$ blocks. The $m_1\text{MCX}$ targets the ancilla q_6 (its controls are q_0, q_1, q_2 in the running example); the $m_2\text{MCX}$ targets the overall target t and never acts on t except as its own target. In particular *no m_1 block ever touches t* .

The $m_1\text{MCX}$ can be decomposed in the following fashion using the toffoli chain technique:

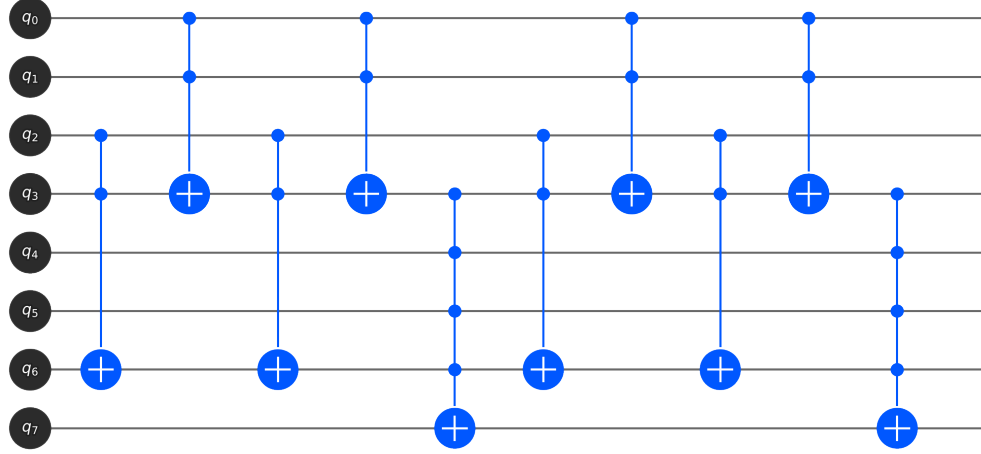


Figure 2: The ancilla-targeting (m_1) MCX gates decomposed (exact version).

If we replace the lowest toffolis of the m_1 MCX block with their approximate versions the circuit is equivalent to the following (consisting of exact toffolis and Δ 's, the diagonal phase debts):

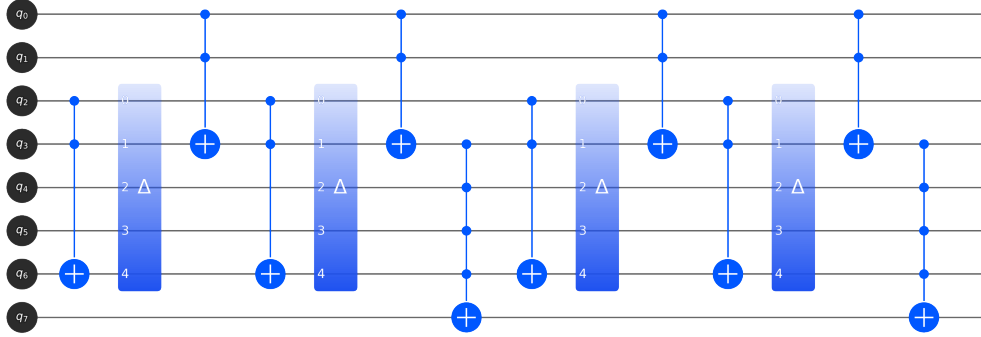


Figure 3: m_1 MCX decomposed: the ancilla-targeting Toffolis replaced with approximate versions.

We want to show that for certain Δ 's (that have a particular activation state), the extra phase debts can be made to cancel.

The key intuition here is that the two m_1 MCX blocks are identical. Since both m_1 MCX blocks are identically constructed, and if they receive the same input, they fire the same amount of diagonal gates Δ .

If the first m_1 MCX fires no Δ , the second will fire no Δ . There will be no residue phase.

If the first m_1 MCX fires one Δ , the second will fire one Δ , and since $\Delta \cdot \Delta = I$ there will be no residue phase.

These intuition will be more rigorously formulated in the following section.

How do we ensure that two m_1 MCX blocks receive the same input? Let's see which qubits matter. q_0 , q_1 , and q_2 are inputs that never gets changed, so we don't have to worry about them. q_3 's value might get flipped in the middle of the m_1 MCX block, but after the last gate in the m_1 MCX block this change is uncomputed, so we can be sure that the second m_1 MCX receives the same value of q_3 as the first. However, the value of q_6 the two blocks receive might be different because the second lower toffoli computes the changed value of q_3 onto q_6 , and there is not a third lower toffoli to uncompute that change.

Hence, if the phase gate gets activated simultaneously when the second lower toffoli in the m_1 MCX ever gets activated, the two m_1 MCX blocks receive different inputs, and there could be a phase residue.

A smart way to work around this problem is to have the phase activate on a very specific activation state – when the second lower toffoli is not activated. For example, consider the phase activation state has $q_2 = 0$ and $q_3 = 1$. Since q_2 is a static control (the control qubit of the lower toffoli that does not get touched by other gates), having the activation state require $q_2 = 0$ means that when the phase could only fire when none of the lower toffolis are active. Since none of the lower toffolis are active, the value q_6 would not get modified after the first m_1 MCX, so both m_1 MCX blocks receive the same input. Therefore, any phase produced by the first m_1 MCX block will be canceled out by the second.

The following sections will more rigorously formulate this idea.

Setup and notation

We use big-endian ordering. Inside its $8k - 6$ decomposition (Fig. 2) the m_1 MCX factors as

$$m_1\text{MCX} = \text{TofA TofB TofA TofB},$$

where TofA are the two Toffolis that *target the ancilla* q_6 (the two that [2]’s $8k - 6$ count keeps exact for this block) and TofB are the rest. We replace each TofA by the same cheap gate

$$\text{TofA}_{\text{approx}} = \Delta \text{TofA},$$

so the same diagonal Δ appears at every occurrence, and we leave the TofB alone. Two pieces of vocabulary describe the controls of TofA:

- the **static control** is the one TofB does *not* touch (here q_2);
- the **dynamic control** is the one TofB *does* touch (here q_3).

The **activation state** $|q_3, q_2\rangle$ is the control pattern on which Δ places its -1 . A relative-phase Toffoli never misbehaves on the fully-on controls (11, where it does its real flip) or the fully-off controls (00); the -1 always sits where exactly one control is on, so the activation state is either $|01\rangle$ or $|10\rangle$. Choosing between those two is the orientation decision that decides everything.

3.3 A bookkeeping tool: separating phase from flip

To track the debt cleanly we want to slide all the stray -1 ’s to one side and keep them apart from the gates’ actual flipping work. Call a unitary *monomial* if it is ΔP with Δ a diagonal of ± 1 ’s (the phase) and P a permutation (the flip). The next two facts are the only tools we need: a phase can be pushed through a flip, and pushing it through just relabels which states it acts on.

Lemma 1. *If Δ is diagonal and P is a permutation, then $\Delta^P := P\Delta P^{-1}$ is diagonal with $(\Delta^P)_j = \Delta_{P^{-1}(j)}$.*

Proof. For any basis state $|j\rangle$,

$$\Delta^P |j\rangle = P\Delta P^{-1} |j\rangle = P\Delta |P^{-1}(j)\rangle = \Delta_{P^{-1}(j)} P |P^{-1}(j)\rangle = \Delta_{P^{-1}(j)} |j\rangle. \quad \square$$

Lemma 2. *Monomial unitaries are closed under multiplication: using $P_1\Delta_2 = \Delta_2^{P_1}P_1$,*

$$(\Delta_1 P_1)(\Delta_2 P_2) = (\Delta_1 \Delta_2^{P_1})(P_1 P_2).$$

With these we can collect the debt of a whole modified block into a single phase sitting in front of the exact gate it was supposed to be.

Lemma 3. *The modified m_1 block B is monomial:*

$$B = \Delta_{\text{net}} (m_1\text{MCX}), \quad \Delta_{\text{net}} = \Delta \Delta^{\text{TofA}'\text{TofB}'},$$

where TofA' , TofB' are the gates padded with identities to a common dimension.

Proof. $\text{TofA}'_{\text{approx}} = \Delta \text{TofA}'$ and $\text{TofB}' = I \cdot \text{TofB}'$ are monomial, so by Lemma 2 their product is too. Pushing both diagonals left,

$$B = \Delta \text{TofA}' \text{TofB}' \Delta \text{TofA}' \text{TofB}' = \Delta \Delta^{\text{TofA}' \text{TofB}'} (\text{TofA}' \text{TofB}' \text{TofA}' \text{TofB}') = \Delta_{\text{net}} (m_1 \text{MCX}). \quad \square$$

So the modified block is exactly the block it was meant to be, times one collected debt Δ_{net} . We just need that debt to vanish in the full circuit.

3.4 The cancellation condition:

Recall the circuit is $BMBM$ with both B 's identical ($M := m_2 \text{MCX}$). For the modified circuit to match the original, the block must, in effect, leave no trace when applied twice — formally, B must be Hermitian, $B^2 = I$. The following lemma makes this an exact equivalence and, crucially, shows the M 's in between do not interfere with the bookkeeping.

Lemma 4. *Let $B_0 := m_1 \text{MCX}$ and $M := m_2 \text{MCX}$, both involutions, and let $U_0 = B_0 M B_0 M = C_{n-2,n}(\sigma_x)$ be the original circuit. With $B = \Delta_{\text{net}} B_0$, the following are equivalent:*

$$(i) BMBM = U_0, \quad (ii) B^2 = I, \quad (iii) \Delta_{\text{net}} \Delta_{\text{net}}^{m_1 \text{MCX}} = I.$$

Proof. (i) is a phase condition. Pushing the inner debt left through $B_0 M$,

$$BMBM = \Delta_{\text{net}} B_0 M \Delta_{\text{net}} B_0 M = \Delta_{\text{net}} \Delta_{\text{net}}^{B_0 M} (B_0 M B_0 M) = \Delta_{\text{net}} \Delta_{\text{net}}^{B_0 M} U_0,$$

so (i) holds iff $\Delta_{\text{net}} \Delta_{\text{net}}^{B_0 M} = I$.

The m_2 blocks drop out. The cheap gates act only on $\{q_2, q_3, q_6\}$, so Δ_{net} is a diagonal that depends on the controls and ancilla but *not on the target t* . Since M permutes basis states by flipping t alone, Lemma 1 gives $M \Delta_{\text{net}} M^\dagger = \Delta_{\text{net}}$, whence

$$\Delta_{\text{net}}^{B_0 M} = B_0 (M \Delta_{\text{net}} M^\dagger) B_0^\dagger = B_0 \Delta_{\text{net}} B_0^\dagger = \Delta_{\text{net}}^{m_1 \text{MCX}},$$

turning the condition into (iii).

(ii) is the same thing. $B^2 = \Delta_{\text{net}} B_0 \Delta_{\text{net}} B_0 = \Delta_{\text{net}} \Delta_{\text{net}}^{m_1 \text{MCX}}$, so $B^2 = I$ iff (iii). All three coincide. $\square$

It pays to read (iii) plainly. With ± 1 entries, it says $(\Delta_{\text{net}})_x (\Delta_{\text{net}})_{m_1 \text{MCX}(x)} = 1$ for every state $|x\rangle$: the debt must take *equal signs* on every pair $\{x, m_1 \text{MCX}(x)\}$. The pair only differs when $m_1 \text{MCX}$ moves x , which happens exactly on

$$S = \{q_0 = q_1 = q_2 = 1\},$$

where it flips the ancilla q_6 .

Hence, for Δ_{net} , its phase activation state cannot be

$$S = \{q_0 = q_1 = q_2 = 1\},$$

. In other words, having the phase fire on $S = \{q_0 = q_1 = q_2 = 1\}$ is a failure mode.

How do we make sure that the phase of Δ_{net} do not fire on $S = \{q_0 = q_1 = q_2 = 1\}$? What does that say about the phase activation state of Δ ?

Notice that because $\Delta_{\text{net}} = \Delta \Delta^{\text{TofA}' \text{TofB}'}$, and Δ only gets activated depending on the values of q_2, q_3 and q_6 , the only important control knob we have is q_2 . We don't have control over q_0, q_1 because Δ does not depend on them. Hence the only thing we can engineer to make sure that failure mode doesn't happen is to set the phase to fire when $q_2 = 0$. If we set the phase activation state to require $q_2 = 0$, we avoid the failure mode $S = \{q_0 = q_1 = q_2 = 1\}$ completely.

Now in the next section we will make this idea more rigorous:

3.5 The right activation state makes the debt vanish

The good choice is to fire the -1 when the *static* control is off. Then the debt lives entirely in the region $q_2 = 0$, while the dangerous region S needs $q_2 = 1$ — the two never meet, so on S there is no debt at all.

Lemma 5 ($|10\rangle$ activation state suffices). *If the activation state is $|q_3, q_2\rangle = |10\rangle$ (static $q_2 = 0$), then $\Delta_{\text{net}} \equiv +1$ on S , so by Lemma 4 the modified circuit equals the original.*

Proof. Every -1 of the bare Δ sits at $q_2 = 0$. We claim the same for $\Delta_{\text{net}} = \Delta \Delta^{\text{TofA'TofB'}}$. Neither padded gate changes q_2 : TofA' targets the ancilla q_6 , and TofB' does not act on the static control q_2 by definition. So TofA'TofB' preserves q_2 on every state, and by Lemma 1 the conjugated factor also carries its -1 's at $q_2 = 0$. Hence all -1 's of Δ_{net} lie in $\{q_2 = 0\}$, disjoint from $S \subseteq \{q_2 = 1\}$. On S we therefore have $\Delta_{\text{net}} = +1$, trivially independent of q_6 . $\square$

The intuition matches the picture exactly: with the static control off at activation, the phase only fires while $q_2 = 0$, but TofA can flip the ancilla only when both its controls are 1 (so $q_2 = 1$). Whenever the debt is in play the ancilla is untouched, both identical blocks see the same input, and the two copies of Δ cancel because $\Delta^2 = I$.

3.6 The wrong activation state leaves a residue

Counterexample for the $|01\rangle$ orientation. Fix the activation state so that the bare phase obeys

$$\Delta_x = -1 \iff (q_6, q_3, q_2) = (1, 0, 1), \quad \Delta_x = +1 \text{ otherwise,}$$

i.e. a -1 exactly when $q_6=1, q_3=0, q_2=1$. By Lemma 4 it suffices to exhibit one pair $\{x, m_1 \text{MCX}(x)\}$ on $S = \{q_0 = q_1 = q_2 = 1\}$ with $(\Delta_{\text{net}})_x (\Delta_{\text{net}})_{m_1 \text{MCX}(x)} = -1$. Recall $(\Delta_{\text{net}})_x = \Delta_x \cdot \Delta_{P^{-1}(x)}$ with $P = \text{TofA'TofB'}$, and that on S the gate TofB flips q_3 , so P^{-1} maps $q_3=0 \leftrightarrow q_3=1$.

Take the q_6 -pair at $q_3 = 0$ on S (kets written big-endian, $|q_6 q_5 q_4 q_3 q_2 q_1 q_0\rangle$, with X a don't-care):

$$x = |0 X X 0 1 1 1\rangle, \quad m_1 \text{MCX}(x) = |1 X X 0 1 1 1\rangle.$$

Their entries in Δ_{net} are

$$\begin{aligned} (\Delta_{\text{net}})_x &= \underbrace{\Delta_x}_{(q_6, q_3, q_2) = (0, 0, 1) \rightarrow +1} \cdot \underbrace{\Delta_{P^{-1}(x)}}_{q_3=1 \rightarrow +1} = +1, \\ (\Delta_{\text{net}})_{m_1 \text{MCX}(x)} &= \underbrace{\Delta_{(1, 0, 1)}}_{(q_6, q_3, q_2) \rightarrow -1} \cdot \underbrace{\Delta_{P^{-1}(m_1 \text{MCX}(x))}}_{q_3=1 \rightarrow +1} = -1. \end{aligned}$$

The pair carries opposite signs, so $\Delta_{\text{net}} \Delta_{\text{net}}^{m_1 \text{MCX}} \neq I$ and, by Lemma 4, the $|01\rangle$ orientation does not yield the original circuit. $\square$

3.7 New Upper Bound:

By Lemma 5 both copies of the ancilla-targeting block may use the $|10\rangle$ cheap gate with no change to the circuit. Each top block has two TofA gates, so four Toffolis are downgraded across the two blocks — exactly the four that [2]'s $8k - 6$ count keeps exact in the ancilla-targeting gates, and the four that [1], Corollary 7.4 already flags as removable. Each downgrade trades a 6-CNOT exact Toffoli for a 3-CNOT cheap one, saving 3 CNOTs, so the four save 12. The target-touching Toffolis of the bottom m_2 blocks must stay exact, so nothing more is available. The bound becomes

$$16n - 40 - 12 = \boxed{16n - 52}.$$

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